Lecture 7 Cash in advance models

MIU: utility depends on real money balances, but never explained why it enters utility.

under certain conditions, equivalent to specifying a transaction tech which involves money.

CIA: focus on transaction demand for money.

Assumption: Consumers must pay for goods in cash

☆ Timing assumption: Svensson (1985) goods market opens before the asset market → agents use cash carried over from last period. ... We follow.

Lucas (1982). agents allocate between cash and other assets after current state realization.

Cash on $\text{hand}_t \geq$ nominal spending on cash goods in t. Here we assume consumption goods are the capital goods

↳ Model setup:

Representative household of mass 1.

no firms, the household operates the technology. Preference:

$$\sum_{t=0}^{\infty} \beta^t U(C_t)$$

$t = 0$ endowments: M_{t-1} units of money

$(1+i_{t-1})B_{t-1}$ units of bond, where i_t is the interest rate

k_{t-1} units of goods

Production Tech: $f(k_{t-1})$ produced, δk_{t-1} after production;

Transaction Tech: C_t purchased with money:

$$M_{t-1} + T_t \geq P_t C_t$$

Government: Costlessly print T_t units of money and lump sum transfer to households

Market prices: goods price P_t, money numeraire 1, bond interest rate i_t

Timing:
1. ① Households enter period t with k_{t-1}, M_{t-1} and $(1+i_{t-1})B_{t-1}$
2. ② Receives transfer T_t from government, thus money at hand:

$$M_{t-1} + T_t$$

3. ③ Produce and sell output for money, sell bonds and receive payoff at the end of period.
4. ④ Use $M_{t-1} + T_t$ to buy goods C_t at price P_t from other households
5. ⑤ Paid for the goods and bonds sold in ③, reach money stock M_t

* Money earned in T can not be used until $T+1$

Thus, we can write out the household's problem.

$$V(W_t, m_{t-1}) = \max_{\{c_t, m_t, k_t, b_t\}} U(c_t) + \beta V(W_{t+1}, m_t)$$ m_t: money W_t: wealth in goods

s.t. ① CIA constraint: $P_t c_t \leq M_{t-1} + T_t \quad \cdots \mu_t$

in real terms $\Longrightarrow \quad c_t \leq \frac{M_{t-1}}{P_t} + \frac{T_t}{P_t} = \frac{M_{t-1}}{P_{t-1}} \frac{P_{t-1}}{P_t} + \frac{T_t}{P_t} = \frac{m_{t-1}}{\pi_t} + \tau_t$

where m_{t-1} is real money balances, $\pi_t = \frac{P_t}{P_{t-1}}$ Inflation. $\tau_t = \frac{T_t}{P_t}$ real lump-sum transfer.

② Budget constraint: $P_t W_t \geq P_t c_t + P_t k_t + M_t + B_t \quad \cdots \lambda_t$

$$P_t W_t = \underbrace{P_t f(k_{t-1})}_{\text{Produced goods}} + \underbrace{P_t (1-\delta) k_{t-1}}_{\text{discounted capital}} + \underbrace{M_{t-1}}_{\text{money}} + \underbrace{T_t}_{\text{transfer}} + \underbrace{(1+i_{t-1}) B_{t-1}}_{\text{bond return}}$$

$$\Rightarrow \quad P_t f(k_{t-1}) + P_t (1-\delta) k_{t-1} + M_{t-1} + T_t + (1+i_{t-1}) B_{t-1} \geq P_t c_t + P_t k_t + M_t + B_t$$

To real terms: $W_t = f(k_{t-1}) + (1-\delta) k_{t-1} + \frac{m_{t-1}}{\pi_t} + \tau_t + (1+i_{t-1}) \frac{b_{t-1}}{\pi_t} \geq c_t + k_t + m_t + b_t$

$$\mathcal{L} = U(c_t) + \beta V(W_{t+1}, m_t) + \lambda_t [W_t - c_t - k_t - m_t - b_t] + \mu_t \left[\frac{m_{t-1}}{\pi_t} + \tau_t - c_t\right]$$

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow U_{c,t} - \lambda_t - \mu_t = 0 \quad ①$$

$$\frac{\partial \mathcal{L}}{\partial k_t} = 0 \Rightarrow \beta V_W (W_{t+1}, m_t) [f'(k_t) + (1-\delta)] - \lambda_t = 0 \quad ②$$

$$\frac{\partial \mathcal{L}}{\partial m_t} = 0 \Rightarrow \beta V_W (W_{t+1}, m_t) \left[\frac{1}{\pi_{t+1}}\right] + \beta V_m (W_{t+1}, m_t) - \lambda_t = 0 \quad ③$$

$$\frac{\partial \mathcal{L}}{\partial b_t} = 0 \Rightarrow \beta V_W (W_{t+1}, m_t) \frac{1+i_t}{\pi_{t+1}} - \lambda_t = 0 \quad ④$$

Envelope theorem: How the optimized value function changes when you change a state/parameter without tracking optimal choices.

$$V(x) = \max_a F(x, a) \Rightarrow V_x(x) = \left.\frac{\partial F(x,a)}{\partial x}\right|_{a = a^*(x)} \quad a^*(x) \text{ solves maximization}$$

Thus, $\underline{V_W (W_t, m_{t-1})} = \lambda_t \quad ⑤$

$\underline{V_m (W_t, m_{t-1})} = \frac{\mu_t}{\pi_t} \quad ⑥$

shadow value of money and wealth (in real terms)

④ & ⑤ $\Rightarrow \quad \lambda_t = \beta \lambda_{t+1} \frac{1+i_t}{\pi_{t+1}} \quad \cdots$ asset pricing euler equation

Case 1: CIA constraint not binding: $\mu_t = 0$.

$V_m (W_t, m_{t-1}) = \frac{\mu_t}{\pi_t} = 0 \quad \cdots$ by Kuhn-Tucker condition

Liquidity service has no value, optimal condition reduced to:

$$\begin{cases} U_{c,t} - \lambda_t = 0 & (7) \\ \beta[f'(k_t) + 1 - \delta]\lambda_{t+1} - \lambda_t = 0 & (8) \\ \beta \frac{1+i_t}{\pi_{t+1}} \lambda_{t+1} = \lambda_t & (9) \text{ ... bond Euler} \\ \frac{\beta}{\pi_{t+1}} \lambda_{t+1} = \lambda_t & (10) \text{ ... money Euler.} \end{cases}$$

$(7) \& (9) \Rightarrow \beta \frac{1+i_t}{\pi_{t+1}} U_{c,t+1} = U_{c,t}$

$(7) \& (8) \Rightarrow \beta[f'(k_t) + 1 - \delta] U_{c,t+1} = U_{c,t}$

real marginal return in capital

$\Rightarrow \underline{f'(k_t) + 1 - \delta} = \frac{1+i_t}{\pi_{t+1}}$

$(9) \& (10) \Rightarrow 1 + i_t = 1 \Rightarrow i_t = 0$

Resembles optimal monetary policy in classical monetary models. (Friedman rule)

bond and money have the same payoff when liquidity has no value, arbitrage $\Rightarrow i_t = 0$

If holding money has no opportunity cost, agents can hold enough cash so that CIA constraint doesn't bind, and money stops distorting choices. $\Rightarrow$ money is irrelevant for real allocations.

Case 2: CIA constraint binding: $\mu_t > 0$ liquidity service has value.

$$U_{c,t} = \lambda_t + \mu_t$$

$$(13) \Rightarrow \beta \lambda_{t+1} \frac{1}{\pi_{t+1}} + \beta \frac{\mu_{t+1}}{\pi_{t+1}} - \lambda_t = 0 \Rightarrow \lambda_t = \beta \frac{\lambda_{t+1} + \mu_{t+1}}{\pi_{t+1}} \quad (11)$$

* Intuition: If you give up a unit of resources today to hold more money, the discounted real payoff equals: ① the real purchase power of that money valued at $\frac{\lambda_{t+1}}{\pi_{t+1}}$ plus ② the liquidity constraint it relaxes in next period's CIA constraint, valued at $\frac{\mu_{t+1}}{\pi_{t+1}}$

$$(11) \Rightarrow \frac{\lambda_t}{P_t} = \beta \left(\frac{\lambda_{t+1}}{P_{t+1}} + \frac{\mu_{t+1}}{P_{t+1}} \right) \quad \text{iterate forward.}$$

The marginal utility value of money today

$$\frac{\lambda_t}{P_t} = \boxed{\sum_{i=1}^{\infty} \beta^i \left(\frac{\mu_{t+i}}{P_{t+i}} \right)}$$ Present discounted value of future liquidity services starting from t+1

$$\frac{\partial V(W_{t+i}, m_{t+i-1})}{\partial M_{t+i-1}} = \frac{\partial V(W_{t+i}, m_{t+i-1})}{\partial m_{t+i-1}} \frac{\partial m_{t+i-1}}{\partial M_{t+i-1}}$$

$$m_{t+i-1} = \frac{M_{t+i-1}}{P_{t+i-1}}$$

$$(6) \Rightarrow \frac{\partial V(W_{t+i}, m_{t+i-1})}{\partial m_{t+i-1}} = \frac{\mu_{t+i}}{\pi_{t+i}}$$

$$\Rightarrow \frac{\partial V(W_{t+i}, m_{t+i-1})}{\partial M_{t+i-1}} = \frac{\mu_{t+i}}{P_{t+i}} \frac{P_{t+i-1}}{P_{t+i-1}} = \frac{\mu_{t+i}}{P_{t+i}}$$

$$\Rightarrow \frac{\lambda_t}{P_t} = \sum_{i=1}^{\infty} \beta^i \frac{\partial V(W_{t+i}, m_{t+i-1})}{\partial M_{t+i-1}}$$ or you can write in in form of derivative w.r.t. real money.

$$\frac{\lambda_t}{P_t} = \sum_{i=1}^{\infty} \beta^i \left(\frac{\mu_{t+i}}{P_{t+i}} \right)$$

$$V_m(W_{t+i}, m_{t+i-1}) = \frac{\mu_{t+i}}{\pi_{t+i}}$$

$$\Rightarrow \frac{\lambda_t}{P_t} = \sum_{i=1}^{\infty} \beta^i \frac{\pi_{t+i} V_m(W_{t+i}, m_{t+i-1})}{P_{t+i}}$$

$$\pi_{t+i} = \frac{P_{t+i}}{P_{t+i-1}}$$

$\Rightarrow$

$$\frac{\lambda_t}{P_t} = \sum_{i=0}^{\infty} \beta^i \frac{V_m(W_{t+i}, m_{t+i-1})}{P_{t+i-1}} = \frac{1}{P_t} \sum_{i=0}^{\infty} \beta^i \frac{V_m(W_{t+i}, m_{t+i-1})}{\frac{P_{t+i-1}}{P_t}}$$

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time discounting $\frac{P_{t+i-1}}{P_t} = \frac{P_{t+i-1}}{P_{t+i-2}} \frac{P_{t+i-2}}{P_{t+i-3}} \cdots \frac{P_{t+1}}{P_t} = \pi_{t+i-1} \cdot \pi_{t+i-2} \cdots \pi_{t+1}$

$$\Rightarrow \lambda_t = \sum_{i=0}^{\infty} \beta^i \frac{\boxed{V_m(W_{t+i}, m_{t+i-1})}}{\prod_{j=1}^{i-1} \pi_{t+j}}$$

→ marginal value of having 1 more unit of predetermined real money balance at beginning of $t+i$ (marginal utility value of liquidity)

future marginal value of money at $t+i-1$ to today's (t) real term.

shadow value of one more real resources today (marginal utility value of wealth), λ_t is the real price of giving up 1 units of goods at time t.

$$\begin{cases} \text{CIA slacks}, & V_m = 0 \ (\mu_t = 0) \\ \text{CIA binds, liquidity scarce} & V_m > 0 \end{cases}$$

* Intuition: Money is an asset whose payoff is not future period consumption itself, but a service flow that can help agents transact more easily.

In the MIU setup: $U(c_t, m_t)$

$$\lambda_t = U_m(c_t, m_t) + \beta E_t\left[\frac{\lambda_{t+1}}{\pi_{t+1}}\right]$$

$$\Rightarrow \frac{\lambda_t}{P_t} = \frac{U_m(c_t, m_t)}{P_t} + \beta E_t\left[\frac{\lambda_{t+1}}{P_{t+1}}\right]$$

Asset pricing recursion: today's price = today's dividend + discounted continuation value.

Forward iterate $\Rightarrow \frac{\lambda_t}{P_t} = E_t \sum_{j=0}^{\infty} \beta^j \frac{U_m(c_{t+j}, m_{t+j})}{P_{t+j}}$

Money is valuable because it yields a stream of utility starting in current period. t

In the CIA setup: $\frac{\lambda_t}{P_t} = \sum_{i=1}^{\infty} \beta^i \left(\frac{\mu_{t+i}}{P_{t+i}}\right) = \sum_{i=1}^{\infty} \beta^i \frac{\partial V(W_{t+i}, m_{t+i-1})}{\partial M_{t+i-1}}$

Money is valuable because it relaxs next period's CIA constraint, starting tomorrow.

§ What is i_t in CIA & MIU?

CIA: Combine (11) & asset pricing euler: $\begin{cases} \lambda_t = \beta \lambda_{t+1} \frac{1+i_t}{\pi_{t+1}} \\ \lambda_t = \beta \frac{\lambda_{t+1} + \mu_{t+1}}{\pi_{t+1}} \end{cases} \Rightarrow \lambda_{t+1} i_t = \mu_{t+1}$

i_t: How valuable liquidity is

MIU: $\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Big\{ U(c_t, m_t) + \lambda_t \Big[f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{m_{t-1}}{\pi_t} + \tau_t + \frac{(1+i_{t-1})b_{t-1}}{\pi_t} - c_t - k_t - m_t - b_t \Big] \Big\}$

MIU's budget constraint: $P_t c_t + P_t k_t + M_t + B_t \le P_t f(k_{t-1}) + P_t(1-\delta)k_{t-1} + M_{t-1} + T_t + (1+i_{t-1})B_{t-1}$

$$\Rightarrow c_t + k_t + \frac{M_t}{P_t} + \frac{B_t}{P_t} \le f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{M_{t-1}}{P_{t-1}}\frac{P_{t-1}}{P_t} + \frac{T_t}{P_t} + (1+i_{t-1})\frac{B_{t-1}}{P_{t-1}}\frac{P_{t-1}}{P_t}$$

to real terms. $\Rightarrow f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{m_{t-1}}{\pi_t} + \tau_t + \frac{(1+i_{t-1})b_{t-1}}{\pi_t} \ge c_t + k_t + m_t + b_t$

$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow U_{c,t} = \lambda_t$ (12) $\quad \frac{\partial \mathcal{L}}{\partial m_t} = 0 \Rightarrow U_{m,t} - \lambda_t + \beta E_t\left[\lambda_{t+1}\frac{1}{\pi_{t+1}}\right] = 0.$

$$\Rightarrow \lambda_t = U_m(c_t, m_t) + \beta E_t\left[\frac{\lambda_{t+1}}{\pi_{t+1}}\right] \quad (13)$$

$$\frac{\partial \mathcal{L}}{\partial b_t} = 0 \Rightarrow \quad -\lambda_t + \beta E_t\left[\lambda_{t+1}\frac{1+i_t}{\pi_{t+1}}\right] = 0$$

$$\Rightarrow \quad \lambda_t = \beta E_t\left[\frac{1+i_t}{\pi_{t+1}}\lambda_{t+1}\right] \quad (14)$$

$$\frac{\partial \mathcal{L}}{\partial k_t} = 0 \Rightarrow \quad -\lambda_t + \beta E_t[\lambda_{t+1}(f'(k_t)+1-\delta)] = 0$$

$$\Rightarrow \quad \lambda_t = \beta E_t[\lambda_{t+1}(f'(k_t)+1-\delta)] \quad (15)$$

$$(14) \Rightarrow \quad \beta E_t\left[\frac{\lambda_{t+1}}{\pi_{t+1}}\right] = \frac{\lambda_t}{1+i_t} \quad \text{plug into } (13) \Rightarrow \lambda_t = U_m(c_t, m_t) + \frac{\lambda_t}{1+i_t}$$

$$\Rightarrow U_m(c_t, m_t) = \lambda_t\left(1 - \frac{1}{1+i_t}\right) \Rightarrow \frac{U_m(c_t, m_t)}{\lambda_t} = \frac{i_t}{1+i_t}$$

direct marginal utility of holding money today

opportunity cost of holding money (money has 0 return, bonds pay i_t, discount to t at $1+i_t$)

(5) Steady state.

From asset pricing euler: $\lambda_t = \beta\lambda_{t+1}\frac{1+i_t}{\pi_{t+1}} \Rightarrow \frac{1+i}{\pi} = \frac{1}{\beta} = R$ ... real interest rate

From (2) capital euler: $\beta V_w(W_{t+1}, m_t)[f'(k_t)+1-\delta] = \lambda_t \Rightarrow \lambda = (f'(k)+1-\delta)\beta V_w$

From (4) bond euler: $\beta V_w(W_{t+1}, m_t)\frac{1+i_t}{\pi_{t+1}} - \lambda_t = 0 \Rightarrow \lambda = \frac{1+i}{\pi}\beta V_w$. $\Rightarrow$

$$f'(k) + (1-\delta) = \frac{1+i}{\pi} = \frac{1}{\beta} \Rightarrow f'(k) = \frac{1}{\beta} - (1-\delta)$$

Steady state capital depends on β and δ, but not π (inflation)

From goods market clearing: $f(k_{t-1}) + (1-\delta)k_{t-1} = c_t + k_t \Rightarrow c = f(k) - \delta k$, consumption also fixed, not dependent on inflation $\Rightarrow$ money superneutrality

Any optimal inflation rate? No, utility is a function of real consumption only. Independent of inflation

However, money superneutrality is not robust to extension of CIA models.

(6) CIA for investment and consumption goods

$$P_t(c_t + I_t) \le M_{t-1} + T_t$$

$$I_t = k_t - (1-\delta)k_{t-1}$$

Then capital euler: $\beta V_w(W_{t+1}, m_t)[f'(k_t)+1-\delta] - \lambda_t - \mu_t + \beta(1-\delta)\mu_{t+1} = 0$

$$V_w(W_{t+1}, m_t) = \lambda_{t+1}$$

$$\Rightarrow \quad \beta[f'(k_t)+1-\delta]\lambda_{t+1} + \beta(1-\delta)\mu_{t+1} = \lambda_t + \mu_t \quad (16)$$

The other FOCs are the same.

S.S. (11) $\Rightarrow \quad \beta\frac{\lambda}{\pi} + \beta\frac{\mu}{\pi} = \lambda \Rightarrow \mu = \left(\frac{\pi}{\beta} - 1\right)\lambda \Rightarrow \frac{\mu}{\lambda} = \frac{\pi}{\beta} - 1$

(16) $\Rightarrow \quad \beta(f'(k) + 1-\delta)\lambda + \beta(1-\delta)\mu = \lambda + \mu \Rightarrow \beta(f'(k)+1-\delta) + \beta(1-\delta)\frac{\mu}{\lambda} = 1 + \frac{\mu}{\lambda}$

$$\Rightarrow \beta(1-\delta) + \beta f'(k) + \beta(1-\delta)\left(\frac{\pi}{\beta} - 1\right) = 1 + \frac{\pi}{\beta} - 1 \Rightarrow \beta f'(k) = \frac{\pi}{\beta} - \pi(1-\delta) \Rightarrow f'(k) = \frac{\pi[1-\beta(1-\delta)]}{\beta^2}$$

Steady state capital depends on inflation rate and is decreasing in π, since $f''(\cdot)<0$

* CIA constraint as a tax on investment, it discourages investment $\Rightarrow$ supernneutrality no long holds

§ CIA with endogenous labor supply. (Cooley and Hansen (1989))

Household's preference:

$$\max_{\{c_t, n_t, m_t, k_t, b_t\}} E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, 1-n_t) = E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_t^{1-\sigma}-1}{1-\sigma} + \psi \frac{(1-n_t)^{1-\eta}}{1-\eta} \right]$$

s.t. $P_t c_t \le M_{t-1} + T_t$ ··· CIA constraint In real terms: $c_t \le \frac{m_{t-1}}{\pi_t} + \tau_t$

$$y_t + (1-\delta)k_{t-1} + \frac{(1+i_{t-1})}{\pi_t} b_{t-1} + \frac{m_{t-1}}{\pi_t} + \tau_t \ge c_t + k_t + b_t + m_t \quad \cdots \text{BC}.$$

Production tech: $y_t = z_t k_{t-1}^{\alpha} n_t^{1-\alpha}$

where $\log z_t = (1-\psi)\log \bar{z} + \psi \log z_{t-1} + \varepsilon_t$

Monetary policy: $M_t = (1+\theta_t) M_{t-1}$

transfer $T_t = \theta_t M_{t-1}$

θ_t is stochastic but known to agent at the beginning of time t:

$$\theta_t = \theta + u_t, \quad u_t \sim AR(1)$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \Big\{ \frac{c_t^{1-\sigma}-1}{1-\sigma} + \psi \frac{(1-n_t)^{1-\eta}}{1-\eta} + \lambda_t \Big[z_t k_{t-1}^{\alpha} n_t^{1-\alpha} - (1-\delta)k_{t-1} + \frac{1+i_{t-1}}{\pi_t} b_{t-1} + \frac{m_{t-1}}{\pi_t} + \tau_t - c_t - k_t - b_t - m_t \Big] + \mu_t \Big[\frac{m_{t-1}}{\pi_t} + \tau_t - c_t \Big]$$

$$\frac{\partial \mathcal{L}}{\partial n_t} = 0 \Rightarrow -\beta^t \psi (1-n_t)^{-\eta} + \beta^t \lambda_t z_t (1-\alpha) k_{t-1}^{\alpha} n_t^{-\alpha} = 0$$

$$\Rightarrow \psi (1-n_t)^{-\eta} = \lambda_t z_t (1-\alpha) k_{t-1}^{\alpha} n_t^{-\alpha}$$

At s.s. $\Rightarrow$ $\psi (1-n)^{-\eta} n^{\alpha} = \lambda \bar{z} (1-\alpha) k^{\alpha}$ (17)

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow \beta^t c_t^{-\sigma} - \lambda_t \beta^t - \mu_t \beta^t = 0 \Rightarrow c_t^{-\sigma} = \lambda_t + \mu_t$$

$$\frac{\partial \mathcal{L}}{\partial m_t} = 0 \Rightarrow \beta^{t+1} \lambda_{t+1} \frac{1}{\pi_{t+1}} - \beta^t \lambda_t + \beta^{t+1} \mu_{t+1} = 0 \Rightarrow \beta \frac{\lambda_{t+1}}{\pi_{t+1}} + \beta \frac{\mu_{t+1}}{\pi_{t+1}} = \lambda_t$$

S.S. $\Rightarrow$ $\lambda = \frac{\beta}{\pi} c^{-\sigma}$ since $\frac{1+i}{\pi} = \frac{1}{\beta} \Rightarrow \frac{\pi}{\beta} = 1+i$

$\lambda = \frac{c^{-\sigma}}{1+i}$ holding c fixed, increasing π, lowers λ, causing RHS of (17) decreasing in π.

(17) LHS increasing in n. Thus, $\pi \uparrow$, $n \downarrow$

Higher inflation is like consumption tax and causes households to substitute toward more leisure.

The effective shadow price of raising consumption is $(1+i)$, i rises 1-1 with s.s. inflation, inflation acts like a consumption tax.